

Atira Todd

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<https://megsgem.github.io/atiratoddportfolio/>

2nd Year Games Programming Student at the University of Staffordshire

Technical Skills:

- Programming Languages: 4 years of C# experience, C++, HLSL, HTML & CSS, Blueprint
- Libraries/APIs: OpenGL, DirectX 11, SDL 2, Monogame
- Game Engines: Unity, Unreal Engine
- Other Software: Git & Github

Projects

- **Physics Simulation Project (C++):**
 - Implemented methods of narrow-phase collision detection and resolution between sphere colliders, axis-aligned bounding boxes and plane colliders
 - Created a rigidbody physics model which would allow forces such as torque to be applied to objects at a given point
 - Used quaternions to allow objects to rotate when a force is applied to them
 - Implemented point mass physics calculations to allow object to move at a constant velocity or acceleration or when a force is applied to them and to allow forces such as gravity, friction and drag to be applied to objects
- **Graphics Application Project (C++, HLSL & DirectX11):**
 - Implemented per-pixel Phong shading by writing shaders using HLSL in order to achieve an accurate lighting model
 - Used inheritance and object oriented principles when creating multiple camera classes which can be stationary or be rotated and used to move around the scene to view it from different angles
 - Used JSON to create configuration files to externally store the objects that would be loaded into the scene including the texture being applied to each one
- **Game Artificial Intelligence Behaviours Project (C# & Unity):**
 - Created various different steering behaviours that could be used by AI agents in games such as a Wander behaviour, Evade and Pursuit behaviours and Flocking behaviours

- Created many pathfinding approaches such as A* and Theta* to allow AI agents to move from one point to another
- Implemented different decision making approaches such as Markov Systems and Finite State Machines

Education

Bsc (Hons) Computer Games Programming, University of Staffordshire

Extracurricular Activities:

- **UKIE Student Games Jam Participant 2025 - After Dark:**
<https://megsgem.itch.io/after-dark>
 - I worked on my own to design and develop a game within a limited 36 hour time frame
 - I used my skills in Unity and C# to program the game's core mechanics and player controls
 - I developed skills in problem solving quickly as well as fast decision making and prototyping in order to complete the game's most important features
- **Bet365 Technical Challenge Hackathon Winner 2025**
 - I worked independently to quickly strategise and find a way to tackle the challenge and to decipher the error codes, broken JSON and scrambled words and decrypt the hidden message
 - I problem solved and worked under time constraints to complete the challenge first in just over an hour
- **University of Staffordshire Computing Society CTF 2026 Winner**
 - I worked independently and quickly to solve the challenges presented to me in under 2 hours
 - I used my current knowledge and the knowledge I acquired during the event on cybersecurity related topics such as steganography and information disclosure to find the hidden flags and complete the challenges
- **University of Staffordshire Games Development Society Member**
- **University of Staffordshire Computing Society Member**